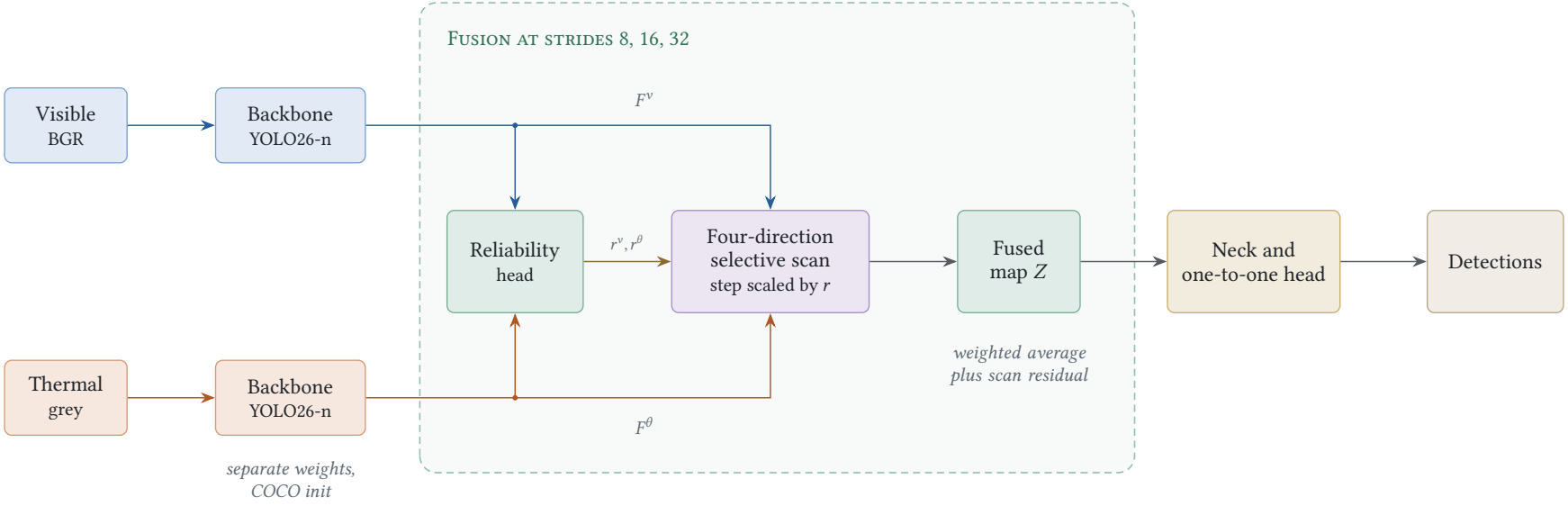

Evaluation

Datasets
LLVIP and M3FD, paired visible–thermal

Matched baseline
concat and 1×1 conv, same backbones, neck, head

Seeds
three seeds for v2 and the baseline

Fault benchmark
11 faults on one camera, 3 severities

Latency
batch one, half precision, 512×640